

Day 54 (2) - Containers

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Container Info

Virtual Machines

- **VMs** allow multiple OS's to run on a single physical server.
 - The OS in each **VM** can be the same or different.
 - **Bins/Libs** are the software libraries/services needed by the apps running in each **VM**.
 - A **VM** allow its app/apps to run in an isolated environment, separate from the apps in the other **VMs**.
 - **VMs** are easy to create, delete, move and etc. They can easily saved and moved between different physical servers.
- A **Hypervisor** is used to manage and allocate hardware resources to each **VM**.
 - A **Type 1 hypervisor** (aka **Native** or **Bare-metal**) is the type of **hypervisor** which runs directly on top of the hardware.
 - They are widely used in data center environments.
 - A **Type 2 hypervisor** (aka **Hosted**) run as a program on an operating system called a **Host OS** like a regular computer program.
 - They can be used to run a virtual network lab on your PC using Cisco Modeling Labs.

Type 1 Hypervisor Visualized	Type 2 Hypervisor Visualized

Containers

Containers are software packages that contain an app and all its required dependencies (**Bins/Libs**) for the contained app to run.

- **Containers** run on a **Container Engine** (ie. **Docker Engine**), which is run on a **host OS**.
- **Containers** are lightweight and include only the dependencies required to run the specific app.
- A **Container Orchestrator** is a software platform automating the deployment, management, scaling etc. of **containers**. These are used to operate large-scale systems (ie. **Microservices**) can require thousands of **containers**.
 - **Kubernetes** is the most popular **container orchestrator**.
 - **Docker Swarm** is Docker's **container orchestration** tool.

Container Visualized

Virtual Machines VS Containers

1. **Virtual Machines:**
 - Can take minutes to boot up as each **VM** runs its own OS.
 - Take up more disk space. (in gigabytes)
 - Use more **CPU/RAM** resources. (each **VM** must run its own OS)
 - Portable and can move between physical systems running the same **hypervisor**.
 - More isolated because each **VM** runs its own OS.
2. **Containers:**
 - Can boot up in milliseconds.
 - Take up very little disk space.
 - Use fewer **CPU/RAM** resources. (Shared OS)
 - Portable as they are smaller, faster to boot up, and Docker **containers** can be run on nearly any **container** service.
 - Less isolated because they all run on the same OS instance; if the OS crashes, all **containers** running on it are effected.